09 Batch Reactors

Batch reactor mole balance:

- In terms of number of moles (*N*): $\frac{dN_A}{dt} = r_A V$. In words, the rate of change in the number of moles of species A changes according to the rate of generation of A times the system volume.
  - If species A is a reactant, the rate of generation of A is negative
  - If species A is a product, the rate of generation of A is positive
  - Hence the balance equation "makes sense" because it properly accounts for the rate of change in the number of moles of the species of interest.
- In terms of conversion (*X*): $\frac{dX_A}{dt} = \frac{-r_A V}{N_{A0}}$
  - Notice that this balance is written with respect to the rate of <u>consumption</u> of A. This makes sense because

    If A is a product, the rate of consumption of A ($-r_A$) is positive, so X_A increases as time goes on.
- Applying the batch reactor mole balances: If you are solving a reactor design problem for a prototypical batch reactor: you can write down the mole balance and just use it without re-deriving it.
  - In other words, we've already derived the mole balance for the prototypical batch reactor. You do not need to rederive it on your problem solutions, except for the following circumstances:
    - The problem statement asks you to derive the balance. In this case, we are wanting to test your ability to derive a balance.
    - The reactor is not a prototypical batch reactor. E.g., it could be a semibatch reactor. In that case, you should derive the appropriate balances for the relevant species, making sure to state your assumptions.
- There are (at least) two types of batch reactors:

Constant volume

*important for gas phase reactions

constant pressure

rigid walls - if the number of moles changes during the reaction, the volume still stays constant

non-rigid walls, e.g., a balloon

if the number of moles changes over the reaction the system volume changes to maintain constant P.

- Let's do an example: The elementary gas phase reaction A + B --> C is carried out isothermally in a constant volume batch reactor and in a constant pressure batch reactor. The initial mixture is equimolar in A and B, and no C is present initially in both cases. Which reactor takes longer to get to 60% conversion?
    - First: Practice with the concept! Pause the video, and try answering <u>conceptually</u>. Why did you choose the answer you did? How confident are you in your response?

Constant volume reactor takes longer!

in the constant P reactor: $\downarrow N_T$ $\downarrow V$ $\uparrow C$ $\uparrow$ rate

    - How would this answer change if this were a liquid phase reaction? in liquid phase, we assume constant volume, so both cases would be the same.
    - Now, let's do the numerical solution:

① constant volume $\rightarrow$ $V = V_0$

$$\frac{dX_A}{dt} = \frac{-r_A V}{N_{A0}} \; . \quad -r_A = k C_A C_B \; .$$

$$C_A = C_{A0}\left(1 - X_A\right) = C_B$$

$$-r_A = k C_{A0}^2 \left(1 - X_A\right)^2$$

$$\frac{dX_A}{dt} = \frac{k C_{A0}^2 (1 - X_A)^2 V}{N_{A0}} \, , \text{ where } C_{A0} = \frac{N_{A0}}{V_0} = \frac{N_{A0}}{V}$$

$$\frac{dX_A}{dt} = k C_{A0} (1 - X_A)^2$$

$$\int_0^{0.6} \frac{dX_A}{(1 - X_A)^2} = k C_{A0} \int_0^t dt$$

$$\boxed{t = \frac{1.5}{k C_{A0}}}$$

constant volume solution

Constant pressure

$$V_0 = \frac{N_{T0} R T_0}{P_0} \qquad V = \frac{N_T R T}{P}$$

$$T = T_0 \, , \quad P = P_0$$

book definition

$$V = V_0 \left(\frac{N_T}{N_{T0}}\right) \longrightarrow \frac{N_T}{N_{T0}} = 1 + \varepsilon X$$

$y_{A0}\,\delta$

$y_{A0} = \frac{1}{2}$, $\varepsilon = \left(\frac{1}{2}\right)(-1)$

mole fraction of the limiting reactant at $t=0$ & δ is the change in the number of moles due to the reaction, per mole of limiting reactant

$$\frac{N_T}{N_{T0}} = 1 - \frac{X_A}{2}$$

$$\underbrace{A+B}_{2\text{ moles}} \rightarrow \underbrace{C}_{1\text{ mole}} \quad \rightarrow \delta = -1$$

- rate law is the same as in the constant V case

$$-r_A = k\, C_A C_B$$

$$C_A = \frac{C_{A0}(1-X_A)}{\left(1-\frac{X_A}{2}\right)} = C_B$$

$$V = V_0\left(1-\frac{X_A}{2}\right)$$

$$\frac{dX_A}{dt} = \frac{-r_A V}{N_{A0}} = \frac{k\, C_{A0}^2 (1-X_A)^2 V}{\left(1-\frac{X_A}{2}\right)^2 N_{A0}}$$

$$\frac{dX_A}{dt} = \frac{k C_{A0}^2 (1-X_A)^2 V_0}{\left(1-\frac{X_A}{2}\right) N_{A0}} = \frac{k C_{A0} (1-X_A)^2}{\left(1-\frac{X_A}{2}\right)}$$

$$\int_0^{0.6} \frac{\left(1-\frac{X_A}{2}\right)}{(1-X_A)^2}\, dX_A = k C_{A0} \int_0^t dt$$

$$\boxed{t = \frac{1.208}{k C_{A0}}}$$

constant pressure case